Algorithms for Big Data

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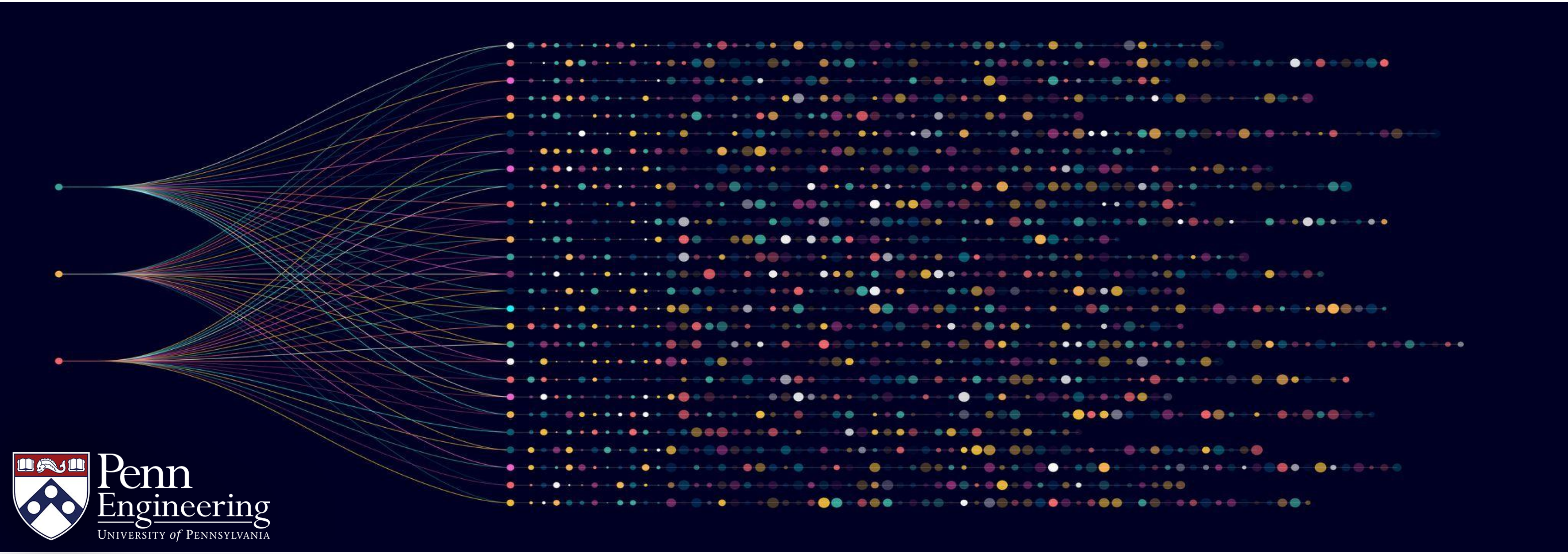
Example Setting of Parameters



Objectives: Computing Large Averages

- Example setting of parameters

- Optimality with respect to δ

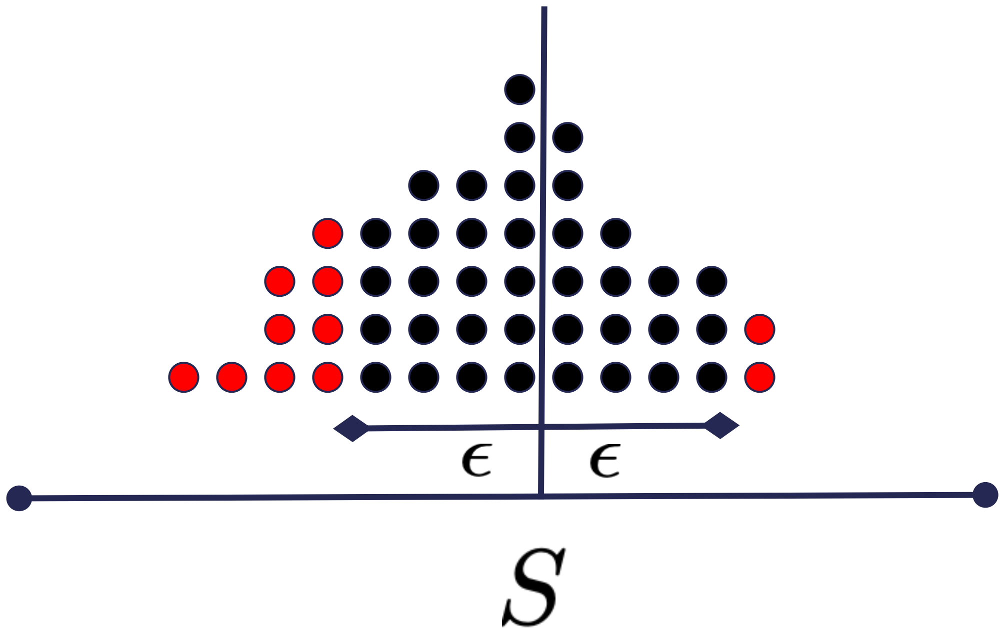
Analysis of the Algorithm

1. Sample $q = \frac{1}{\delta\epsilon^2}$ positions:

$$\boldsymbol{i}_1, \ldots, \boldsymbol{i}_q \sim [n]$$

2. Output empirical average:

$$\hat{\boldsymbol{S}} = \frac{1}{q}\sum_{t=1}^{q} X_{\boldsymbol{i}_t}$$



$$\Pr\left[\left|\hat{\boldsymbol{S}} - S\right| \geq \epsilon\right] \leq \delta$$

How many users are in Europe?

- Total number of Facebook users: ~3 billion

- For accuracy parameter 0.01, and failure probability 0.1, it suffices to consider 100,000 queries.
  - That’s only 0.003% of the dataset!

- The output is correct 90% of the time. If it is correct, the fraction of European users is $\hat{S}$ up to plus or minus 0.01.

How many users are in Europe?

- Total number of Facebook users: ~3 billion

- For accuracy parameter ~~0.01~~ 0.05, and failure probability 0.1, it suffices to consider ~~100,000~~ 4,000 queries.

- The output is correct 90% of the time. If it is correct, the fraction of European users is $\hat{S}$ up to plus or minus ~~0.01~~ 0.05

Is it optimal?

- Two parameters: accuracy and failure probability
- Query complexity: $q = \dfrac{1}{\delta \epsilon^2}$
- Dependence on ϵ is optimal
- Dependence on δ can be improved $O\left(\dfrac{\log(1/\delta)}{\epsilon^2}\right)$